

Database Tuning

Normalization

Normal Forms and Normalizations



Anomalies 1 2 3 4 5 | Normal Forms

Employees

The Employee Table

The table **employee** record the salaries of the different employees in our organization. An agreement with the trade unions imposes that salaries are determined by the position. The actual value has been negotiated and fixed. The salary of a clerk is \$2000 per month, the salary of a manager is \$3000 per month, *etc.*

Employees

number	name	department	position	salary
1XU3	Dewi Srijaya	sales	clerk	2000
5CT4	Axel Bayer	marketing	trainee	1200
4XR2	John Smith	accounting	clerk	2000
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Unenforced Functional Dependencies

The functional dependency is violated. Two managers have different salaries.

We cannot enforce by making **position** as the **primary key**.

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Redundancy

With this design, information is repeated.

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NULL

We cannot remember the salary of an engineer unless we have at least one engineer.

Alternatively, we can use **NULL**. However, if **number** is the **primary key**, then we cannot even use **NULL**.

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Deletion Anomaly

We cannot accept **Axel Bayer**'s resignation from trainee position. If not we would forget the salary of a trainee if Axel leaves.

FD $\leadsto$ PK / CK
 / \
 unique not null

Anomalies | **Normal Forms** ❶ ❷ ❸ ❹ ❺ ❻ ❼

NF?

Idea #1

Let us require that every attribute **fully depends*** on the primary key.

*Y **fully depends** on X in $X \rightarrow Y$ if Y is not dependent on any proper subset of X.

Anomalies | Normal Forms 1 2 3 4 5 6 7

NF?

Idea #1

Let us require that every attribute **fully depends*** on the primary key.

Issue

 $\{A, B\}$ $A \rightarrow A$

But **prime attributes** (*attributes in a candidate key*) do not fully depend on the key (*if there is a composite key*).

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Anomalies | **Normal Forms** ❶ ❷ ❸ ❹ ❺ ❻ ❼

NF?

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Let us require that every attribute **fully depends*** on the primary key.

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But **prime attributes** (*attributes in a candidate key*) do not fully depend on the key (*if there is a composite key*).

Idea #1 (refined)

Let us require that every **non-prime** attribute fully depends* on the primary key.

*Y **fully depends** on X in $X \rightarrow Y$ if Y is not dependent on any proper subset of X.

Anomalies | **Normal Forms** ❶ ❷ ❸ ❹ ❺ ❻ ❼

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But there could be **more than one** candidate key!

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Anomalies | Normal Forms 1 2 3 4 5 6 7

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Idea #1 (refined)

Let us require that every **non-prime** attribute fully depends* on the primary key.

Issue

But there could be **more than one** candidate key!

Idea #1 (refined again)

Let us require that every **non-prime** attribute fully depends* on **each candidate key**.

*Y **fully depends** on X in $X \rightarrow Y$ if Y is not dependent on any proper subset of X.

2NF

Definition / Theorem

A relation R with a set of functional dependencies Σ is in **Second Normal Form (2NF)** if for every functional dependency $X \rightarrow \{A\} \in \Sigma^+$.

- $X \rightarrow \{A\}$ is **trivial**, or
- A is a **prime attribute**, or
- X is **not** a **proper subset** of a candidate key.

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Not in 2NF

A relation R with a set of functional dependencies Σ is **NOT** in **Second Normal Form (2NF)** if there is a functional dependency $X \rightarrow \{A\} \in \Sigma^+$.

- $X \rightarrow \{A\}$ is **non-trivial**, and
- A is **not** a **prime attribute**, and
- X is a **proper subset** of a candidate key.

Anomalies | Normal Forms 1 2 3 4 5 6 7

NF?

Idea #2

Let us require that no attribute is **transitively dependent*** on any candidate key.

Why?

What is the issue with **transitive dependence**?

*Z is **transitively dependent** on X in $X \rightarrow Z$ if there is another attribute Y that is **not** a candidate key such that $X \rightarrow Y$ and $Y \rightarrow Z$.

BCNF

Definition / Theorem

Edgar F. Codd

A relation R with a set of functional dependencies Σ is in **Boyce-Codd Normal Form (BCNF)** if for every functional dependency $X \rightarrow \{A\} \in \Sigma^+$.

- $X \rightarrow \{A\}$ is **trivial**, or
- X is a **superkey**.

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Anomalies | Normal Forms 1 2 3 4 5 6 7

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Question

What about the following case?

$$R = \{A, B, C\}$$
$$\Sigma = \{ \{A, B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\} \}$$

Anomalies | Normal Forms 1 2 3 4 5 6 7

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- $X \rightarrow \{A\}$ is **trivial**, or
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Question

What about the following case?

$R = \{A, B, C\}$

$\Sigma = \{ \{A, B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\} \}$



Issue

- $\{C\} \rightarrow \{B\}$ is non-trivial.
- $\{C\}$ is not a superkey.

It is **not** in BCNF.

Anomalies | **Normal Forms** 1 2 3 4 5 6 7

NF?

Idea #3

Let us relax the BCNF requirements for prime attributes in order to cater for such cases.

3NF

Definition / Theorem

A relation R with a set of functional dependencies Σ is in **Third Normal Form (3NF)** if for every functional dependency $X \rightarrow \{A\} \in \Sigma^+$.

- $X \rightarrow \{A\}$ is **trivial**, or
- A is a **prime attribute**, or
- X is a **superkey**.

Not in 3NF

A relation R with a set of functional dependencies Σ is **NOT** in **Third Normal Form (3NF)** if there is a functional dependency $X \rightarrow \{A\} \in \Sigma^+$.

- $X \rightarrow \{A\}$ is **non-trivial**, and
- A is **not** a **prime attribute**, and
- X is **not** a **superkey**.

$$2NF \supset 3NF \supset BCNF$$

Break



Introduction | Decomposition

Methods

Normalization Methods algo to make fragment each in NF

The three common methods for relational database schema design are the **decomposition** method, the **synthesis** method, and the **entity-relationship** approach.

Introduction ⓘ ⓘ ⓘ | Decomposition

Methods

Normalization Methods

The three common methods for relational database schema design are the **decomposition** method, the **synthesis** method, and the **entity-relationship** approach.

Decomposition Method

The **decomposition** method is based on the assumption that a database can be represented by a universal relation* (i.e., *universal relation assumption*).

This relation is then decomposed into smaller relations called fragments, in order to remove redundant data.

*There is a relation that contains all the attributes of the database.

Introduction ⓘ ⓘ ⓘ | Decomposition

Methods

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Decomposition Method

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This relation is then decomposed into smaller relations called fragments, in order to remove redundant data.

Synthesis Method

The **synthesis** method is based on the assumption that a database can be described a given set of attributes and a given set of functional dependencies.

The 3NF (or *occasionally BCNF*) relations fragments are then synthesized based on the given set of dependencies.

*There is a relation that contains all the attributes of the database.

Introduction | **Decomposition** ❶ ❷ ❸ ❹ ❺ ❻ ❼ ❽ ❾ ❿

Criteria

Two Criteria

The two **main** criteria for the decomposition and synthesis methods are losslessness (*reconstructability*) and dependency_preserving. (*covering*).

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

Criteria

Two Criteria

The two **main** criteria for the decomposition and synthesis methods are losslessness (*reconstructability*) and dependency preserving (*covering*).

Losslessness

A decomposition is a **lossless-join** decomposition if the **natural join*** of all the fragments is equivalent to the original relation.

$$R = R_1 \bowtie R_2 \bowtie \dots \bowtie R_n$$

$$R \subseteq \underline{R_1 \bowtie R_2}$$

This a **join dependency**.

*See the universal relation assumption, assume no NULL values.

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

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Dependency Preservation

A decomposition is a **dependency preserving** decomposition if the information captured by the functional dependencies are captured. In other words, the union of the **projected sets functional dependencies** is equivalent to the original set Σ

$$\Sigma \equiv (\Sigma|_{R_1} \cup \Sigma|_{R_2} \cup \dots \cup \Sigma|_{R_n})$$

*See the universal relation assumption, assume no **NULL** values.

Losslessness

Lossless-Join Decomposition

If we take into account the **NULL** values, the decomposition is lossless if the **full outer join** of the two tables on the condition that their primary and foreign keys are equal **reconstitutes** the initial table.

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1XU3	Dewi Srijaya	sales	clerk	2000
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```
INSERT INTO salaries ( SELECT DISTINCT position, salary FROM employees );  
-- alter table to remove `salary` from Employees and removes duplicates,  
--      set the foreign key, clean `NULL`, etc
```

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Employees $n_o \rightarrow n, d, p$

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Salaries $p \rightarrow s$

position	salary
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```
SELECT * FROM employees e FULL OUTER JOIN salaries s
WHERE e.position = s.position;
```

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Losslessness

Lemma 1: Binary Decomposition

A binary decomposition of R is a pair of tables $\delta = \{R_1, R_2\}$ such that

$$R = R_1 \cup R_2$$

It is a lossless-join decomposition if

$$(R_1 \cap R_2) \rightarrow R_1 \quad \underline{\text{or}} \quad (R_1 \cap R_2) \rightarrow R_2$$

Lemma 2: General Decomposition

A decomposition of R into $R_1, R_2, \dots, R_n$ is a lossless-join decomposition if there exists at least one sequence of binary lossless-join decomposition that generates the decomposition.



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Losslessness

The Chase for Losslessness

1. Create a table r with schema R with n tuples.
2. For each $A \in R_i$, **distinguish** the A -values in the i^{th} tuple (*i.e., set as α*).
3. Chase until you find a **row of distinguished variables**.

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Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C\}, \{A, C\} \rightarrow \{B, D\}, \{C\} \rightarrow \{A, B\}, \{D\} \rightarrow \{A, E\}, \{B, D\} \rightarrow \{A, C\} \}$$

Is $\{ \{A, B, C\}, \{B, C, D\}, \{B, D, E\} \}$ following decomposition a lossless decomposition?

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A	B	C	D	E
α	α	α	d_1	e_1
a_2	α	α	α	e_2
a_3	α	c_3	α	α

ABC

BCD

BDE

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Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

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α	α	c_3	α	α

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 $R = \{A, B, C, D, E\}$ $\Sigma = \{ \{A,B\} \rightarrow \{C\} , \{A,C\} \rightarrow \{B,D\} , \{C\} \rightarrow \{A,B\} , \{D\} \rightarrow \{A,E\} , \{B,D\} \rightarrow \{A,C\} \}$ Is $\{ \{A,B,C\}, \{B,C,D\}, \{B,D,E\} \}$ following decomposition a lossless decomposition? 

A	B	C	D	E
α	α	α	d_1	e_1
α	α	α	α	α
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Is $\{ \{A, B, C\}, \{B, C, D\}, \{A, E\} \}$ following decomposition a lossless decomposition? X

A	B	C	D	E
α	α	α	α	1
α	α	α	α	1
α	3	3	3	α

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

Projection

Projected Functional Dependencies

Consider a relation R with a set of functional dependencies Σ . A set Σ' of **projected functional dependencies** on R' from R with Σ , where $R' \subseteq R$, is the set of functional dependencies equivalent to the set of functional dependencies $X \rightarrow Y$ in Σ^+ such that $X \subseteq R'$ and $Y \subseteq R'$.

In Other Words

- $X \subseteq R'$: All attributes in X comes from R' .
- $Y \subseteq R'$: All attributes in Y comes from R' .
- $X \rightarrow Y$ in Σ^+ : The functional dependencies can be logically derived from Σ .

$$\Sigma|_{R_i} \quad \Sigma|_R = ?$$

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$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

$$\{B\} \rightarrow \{D\}$$

What is the set of **projected functional dependencies** Σ' on $R' = \{A, B, D, E\}$ from R with Σ ?

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Question

Let

$$R = \{A, B, C, D, E\}$$

$$\Sigma = \{ \{A, B\} \rightarrow \{C, D, E\}, \{A, C\} \rightarrow \{B, D, E\}, \{B\} \rightarrow \{C\}, \{C\} \rightarrow \{B\}, \{C\} \rightarrow \{D\}, \{B\} \rightarrow \{E\}, \{C\} \rightarrow \{E\} \}$$

What is the set of **projected functional dependencies** Σ' on $R' = \{A, B, D, E\}$ from R with Σ ?

Answer?

$$\{ \{A, B\} \rightarrow \{D, E\}, \{B\} \rightarrow \{D, E\} \}$$

Preservation

Dependency Preservation

A decomposition is a **dependency preserving** decomposition if the information captured by the functional dependencies are captured. In other words, the union of the **projected sets functional dependencies** is **equivalent** to the original set Σ

$$\Sigma \equiv (\Sigma|_{R_1} \cup \Sigma|_{R_2} \cup \dots \cup \Sigma|_{R_n})$$

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Salaries

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Question

Is this a dependency preserving decomposition?

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

Preservation

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$$\Sigma \equiv (\Sigma|_{R_1} \cup \Sigma|_{R_2} \cup \dots \cup \Sigma|_{R_n})$$

Employees

$\Sigma = \{$
 $\{\text{number}\} \xrightarrow{+} \{\text{name}, \text{department}, \text{position}, \text{salary}\},$
 $\{\text{position}\} \xrightarrow{+} \{\text{salary}\}$
 $\}$

Salaries

$\Sigma|_{R_1} = \{ \{\text{number}\} \rightarrow \{\text{name}, \text{department}, \text{position}\} \}$
 $\Sigma|_{R_2} = \{ \{\text{position}\} \rightarrow \{\text{salary}\} \}$

$$\Sigma \equiv \Sigma|_{R_1} \cup \Sigma|_{R_2}$$

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

Algorithm

Algorithm 3: Decomposition Algorithm**input:** R, Σ **output:** δ **if** (R is **not** in the required normal form) **then**Let $X \rightarrow Y$ be a violation;**Decompose** R into two relations: $R_1 := X^+$ (with respect to Σ) $R_2 := (R - X^+) \cup X$ (with respect to Σ)**return** { $\text{Algo3}(R_1, \Sigma|_{R_1}), \text{Algo3}(R_2, \Sigma|_{R_2})$ };**else****return** { R };

Algorithm

Algorithm 3: Decomposition Algorithm

input: R, Σ **output:** δ **if** (R is **not** in the required normal form) **then**Let $X \rightarrow Y$ be a violation;**Decompose** R into two relations: $R_1 := X^+$ (with respect to Σ) $R_2 := (R - X^+) \cup X$ (with respect to Σ)**return** { Algo3($R_1, \Sigma|_{R_1}$), Algo3($R_2, \Sigma|_{R_2}$) };**else****return** { R };

Properties

The decomposition algorithm

- works for many normal forms, and
- always terminates, and
- guaranteed to find **lossless-join decomposition**.

$$R_1 \cap R_2 = X$$

$$X \rightarrow X^+ \text{ (by definition)}$$

$$(R_1 \cap R_2) \rightarrow R_1$$

Hence, lossless-join (by lemma 1)

However, the decomposition may **not** be **dependency preserving**.

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

Algorithm

Algorithm 4: Synthesis Algorithm**input:** R, Σ **output:** δ $\Sigma^- = \text{Algo2}(\Sigma);$ // *minimal cover* $\delta = \emptyset$ // *start with empty***for each** $(X \rightarrow Y \in \Sigma^-)$ **do****if** (there is **no** $R_i \in \delta$ such that $(X \cup Y) \subseteq R_i$) **then**// *i.e., not subsumed* $\delta = \delta \cup (X \cup Y);$ // *construct a fragment***if** (there is **no** $R_i \in \delta$ that contain one of the keys) **then**Add one key to the δ **return** $\delta;$

Algorithm

Algorithm 4: Synthesis Algorithm**input:** R, Σ **output:** δ $\Sigma^- = \text{Algo2}(\Sigma);$ // minimal cover $\delta = \emptyset$ // start with empty**for each** $(X \rightarrow Y \in \Sigma^-)$ **do** **if** (there is **no** $R_i \in \delta$ such that $(X \cup Y) \subseteq R_i$) **then**

// i.e., not subsumed

 $\delta = \delta \cup (X \cup Y);$ // construct a fragment **if** (there is **no** $R_i \in \delta$ that contain one of the keys) **then** Add one key to the δ **return** $\delta;$ **Properties**

The synthesis algorithm

- works for 3NF (and may find BCNF), and
- always terminates, and
- guaranteed to find **lossless-join decomposition**, and
this is handled by adding the key
- guaranteed to find **dependency preserving decomposition**.

Fragments R_i are from $(X \cup Y)$ $R_i \models X \rightarrow Y$ (from Σ^-)

Since this is done for all $X \rightarrow Y \in \Sigma^-$, we can reconstruct Σ^-

But $\Sigma^- \equiv \Sigma$

Hence, dependency preserving

Introduction | **Decomposition** 1 2 3 4 5 6 7 8 9 10

Algorithm

Algorithm 4: Synthesis Algorithm**input:** R, Σ **output:** δ $\Sigma^- = \text{Algo2}(\Sigma);$ // minimal cover $\delta = \emptyset$ // start with empty**for each** $(X \rightarrow Y \in \Sigma^-)$ **do** **if** (there is **no** $R_i \in \delta$ such that $(X \cup Y) \subseteq R_i$) **then**

// i.e., not subsumed

 $\delta = \delta \cup (X \cup Y);$ // construct a fragment **if** (there is **no** $R_i \in \delta$ that contain one of the keys) **then** Add one key to the δ **return** $\delta;$ **Improvement**

We should always use **canonical cover** instead of minimal cover to reduce the number of tables created.

Properties

The synthesis algorithm

- works for 3NF (*and may find BCNF*), and
- always terminates, and
- guaranteed to find **lossless-join decomposition**, and
this is handled by adding the key
- guaranteed to find **dependency preserving decomposition**.

Fragments R_i are from $(X \cup Y)$ $R_i \models X \rightarrow Y$ (*from Σ^-*)

Since this is done for all $X \rightarrow Y \in \Sigma^-$, we can reconstruct Σ^-

But $\Sigma^- \equiv \Sigma$

Hence, dependency preserving

Case • @ | BCNF | 3NF | Schema

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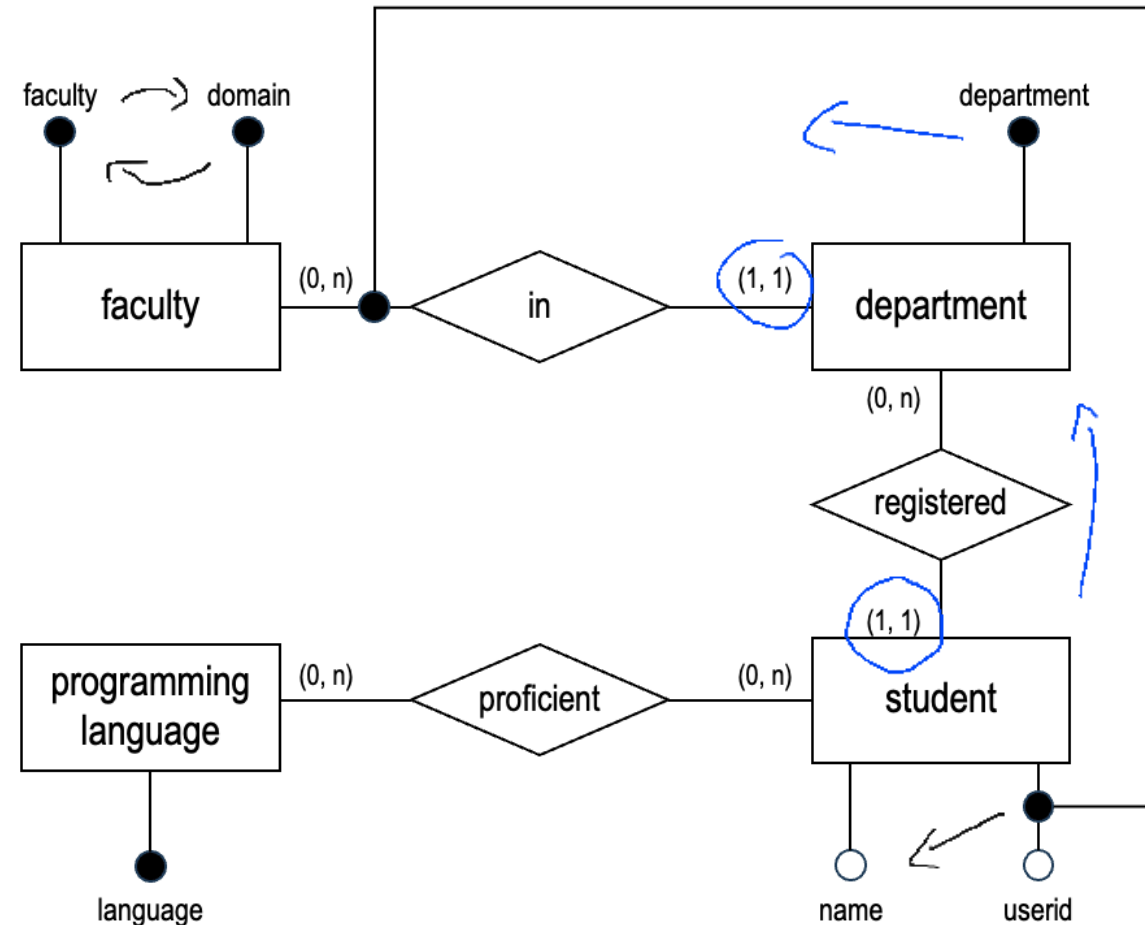
Question

Consider the following entity-relationship diagram.

- Determine the set of functional dependencies Σ that holds on the schema $R = \{A, B, C, D, E, F\}$.
- Find the **best*** decomposition.

Best here is either BCNF or 3NF, depending on which can be dependency preserving.

Attribute	Letter	Attribute	Letter
name	A	department	D
userid	B	faculty	E
domain	C	language	F



Case | BCNF | 3NF | Schema

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Question

Consider the following entity-relationship diagram.

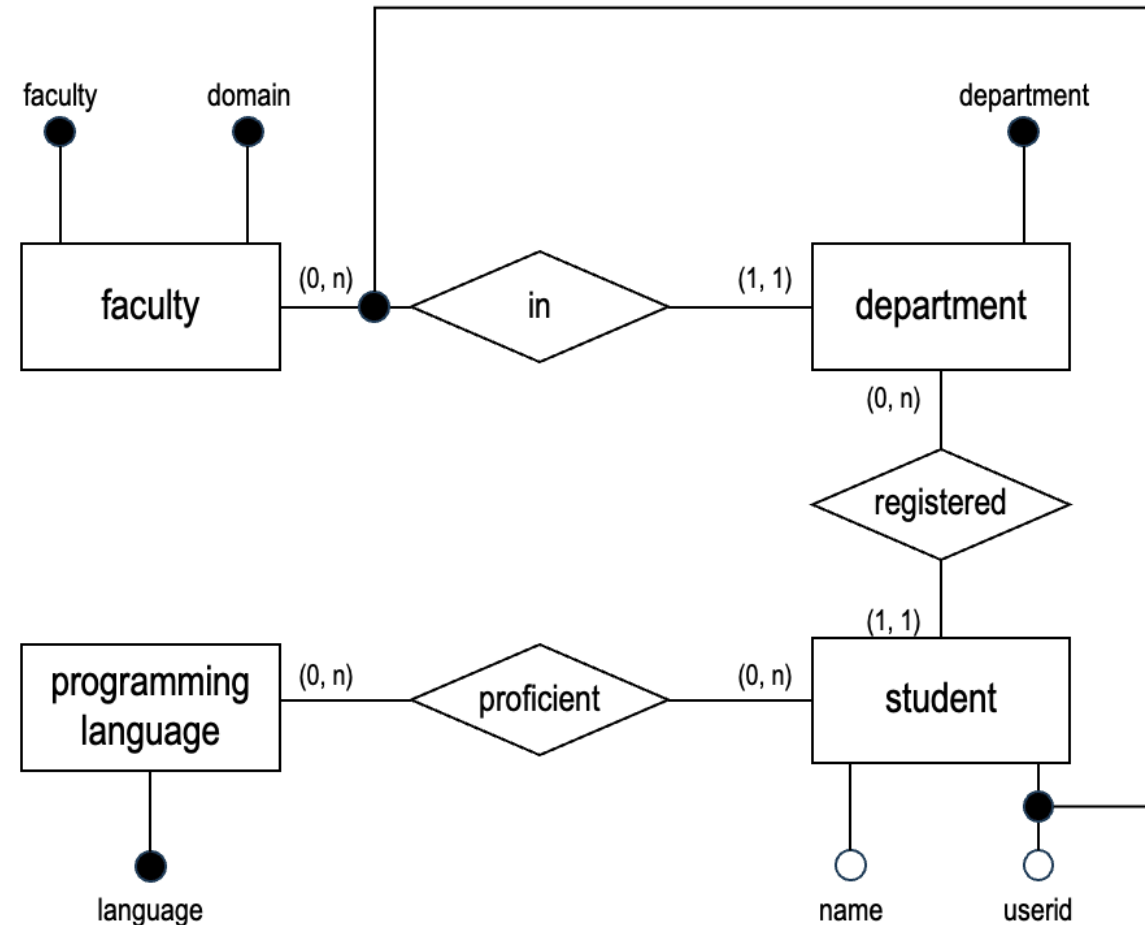
- Determine the set of functional dependencies Σ that holds on the schema $R = \{A, B, C, D, E, F\}$.
- Find the **best*** decomposition.

Best here is either BCNF or 3NF, depending on which can be dependency preserving.

$\Sigma = \{$

$\{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\},$
 $\{D\} \rightarrow \{C, E\},$
 $\{B, C\} \rightarrow \{A, D\}, \{B, E\} \rightarrow \{A, D\}$

$\}$



Case | **BCNF** 1 2 3 4 5 6 | 3NF | Schema

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BCNF Decomposition

BCF BDF BEF

Let $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C, E\}, \{B, C\} \rightarrow \{A, D\}, \{B, E\} \rightarrow \{A, D\} \}$ Find a BCNF decomposition of R with Σ .

Case | **BCNF** 1 2 3 4 5 6 | 3NF | Schema

Sentosa University of Technology

BCNF DecompositionLet $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C, E\}, \{B, C\} \rightarrow \{A, D\}, \{B, E\} \rightarrow \{A, D\} \}$ Find a BCNF decomposition of R with Σ .**Using $\{C\} \rightarrow \{E\}$ on $\{A, B, C, D, E, F\}$**

- $\{C\}^+ = \{C, E\}$
- Decompose $\{A, B, C, D, E, F\}$ into
 - **$\{C, E\}$** with $\Sigma|_{\{C, E\}} = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\} \}$
 - **$\{A, B, C, D, F\}$** with $\Sigma|_{\{A, B, C, D, F\}} = \{ \{B, C\} \rightarrow \{A, D\}, \{B, D\} \rightarrow \{A, C\}, \{D\} \rightarrow \{C\} \}$

Case | **BCNF** 1 2 3 4 5 6 | 3NF | Schema

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BCNF DecompositionLet $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C, E\}, \{B, C\} \rightarrow \{A, D\}, \{B, E\} \rightarrow \{A, D\} \}$ Find a BCNF decomposition of R with Σ .**Using $\{C\} \rightarrow \{E\}$ on $\{A, B, C, D, E, F\}$**

- Decompose $\{A, B, C, D, E, F\}$ into $\{C, E\}$ and $\{A, B, C, D, F\}$

 $\{C, E\}$ is in BCNF**Using $\{D\} \rightarrow \{C\}$ on $\{A, B, C, D, F\}$**

- $\{D\}^+ = \{C, D\}$
- Decompose $\{A, B, C, D, F\}$ into
 - $\{C, D\}$ with $\Sigma|_{\{C, D\}} = \{ \{C\} \rightarrow \{D\} \}$
 - $\{A, B, D, F\}$ with $\Sigma|_{\{A, B, D, F\}} = \{ \{B, D\} \rightarrow \{A\} \}$

Case | **BCNF** 1 2 3 4 5 6 | 3NF | Schema

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BCNF DecompositionLet $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C, E\}, \{B, C\} \rightarrow \{A, D\}, \{B, E\} \rightarrow \{A, D\} \}$ Find a BCNF decomposition of R with Σ .**Using $\{C\} \rightarrow \{E\}$ on $\{A, B, C, D, E, F\}$**

- Decompose $\{A, B, C, D, E, F\}$ into $\{C, E\}$ and $\{A, B, C, D, F\}$

 $\{C, E\}$ is in BCNF**Using $\{D\} \rightarrow \{C\}$ on $\{A, B, C, D, F\}$**

- Decompose $\{A, B, C, D, F\}$ into $\{C, D\}$ and $\{A, B, D, F\}$

 $\{C, D\}$ is in BCNF**Using $\{B, C\} \rightarrow \{A\}$ on $\{A, B, D, F\}$**

- $\{B, D\}^+ = \{A, B, D\}$
- Decompose $\{A, B, D, F\}$ into
 - $\{A, B, D\}$ with $\Sigma|_{\{A, B, D\}} = \{ \{B, D\} \rightarrow \{A\} \}$
 - $\{B, D, F\}$ with $\Sigma|_{\{B, C, F\}} = \emptyset$

Case | **BCNF** 1 2 3 4 5 6 | 3NF | Schema

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BCNF DecompositionLet $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C, E\}, \{B, C\} \rightarrow \{A, D\}, \{B, E\} \rightarrow \{A, D\} \}$ Find a BCNF decomposition of R with Σ .**Using $\{C\} \rightarrow \{E\}$ on $\{A, B, C, D, E, F\}$**

- Decompose $\{A, B, C, D, E, F\}$ into $\{C, E\}$ and $\{A, B, C, D, F\}$

 $\{C, E\}$ is in BCNF**Using $\{D\} \rightarrow \{C\}$ on $\{A, B, C, D, F\}$**

- Decompose $\{A, B, C, D, F\}$ into $\{C, D\}$ and $\{A, B, D, F\}$

 $\{C, D\}$ is in BCNF**Using $\{B, C\} \rightarrow \{A\}$ on $\{A, B, D, F\}$**

- Decompose $\{A, B, D, F\}$ into $\{A, B, D\}$ and $\{B, D, F\}$

 $\{A, B, D\}$ is in BCNF **$\{B, D, F\}$ is in BCNF**

Case | **BCNF** 1 2 3 4 5 6 | 3NF | Schema

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Fragments

- {C,E} { {C} → {E}, {E} → {C} }
- {C,D} { {C} → {D}, {D} → {C} }
- {A,B,D} { {B,D} → {A} }
- {B,D,F} ∅

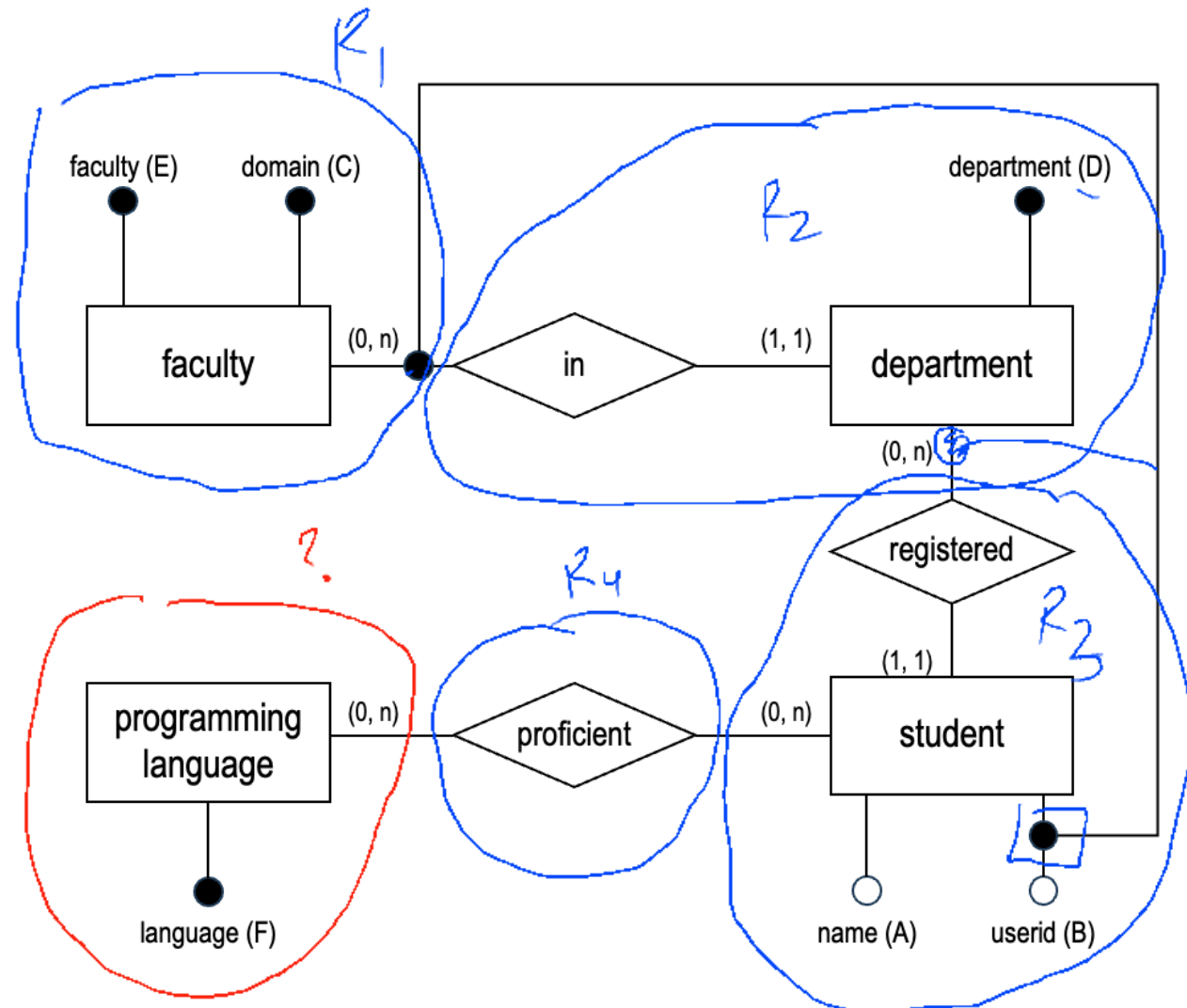
$\Sigma = \{$

{C} → {E}, {E} → {C},
 {D} → {C,E},
 {B,C} → {A,D}, {B,E} → {A,D}

$\}$

Question

Is it dependency preserving?



Case | BCNF | **3NF** 1 2 3 4 5 | Schema

Sentosa University of Technology

3NF Decomposition

Let $R = \{A, B, C, D, E, F\}$

$\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C,E\}, \{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\} \}$

Find a 3NF decomposition of R with Σ .

Case | BCNF | **3NF** 1 2 3 4 5 | Schema

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3NF Decomposition

Let $R = \{A, B, C, D, E, F\}$

$\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C,E\}, \{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\} \}$

Find a 3NF decomposition of R with Σ .

Initial Step

Keys: $\{B,C,D\}, \{B,C,F\}, \{B,E,F\}$

Canonical Cover: $\{ \{B,E\} \rightarrow \{A,D\}, \{C\} \rightarrow \{E\}, \{D\} \rightarrow \{E\}, \{E\} \rightarrow \{C\} \}$

Case | BCNF | **3NF** 1 2 3 4 5 | Schema

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3NF DecompositionLet $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C,E\}, \{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\} \}$ Find a 3NF decomposition of R with Σ .**Initial Step**Keys: $\{B,C,D\}, \{B,C,F\}, \{B,E,F\}$ Canonical Cover: $\{ \{B,E\} \rightarrow \{A,D\}, \{C\} \rightarrow \{E\}, \{D\} \rightarrow \{E\}, \{E\} \rightarrow \{C\} \}$ **Construction** $\{B,E\} \rightarrow \{A,D\}$

ABDE

 $\{C\} \rightarrow \{E\}$

CE

 $\{D\} \rightarrow \{E\}$ ~~DE~~ $\{E\} \rightarrow \{C\}$ ~~CE~~

Case | BCNF | **3NF** 1 2 3 4 5 | Schema

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3NF Decomposition

Let $R = \{A, B, C, D, E, F\}$

$\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C,E\}, \{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\} \}$

Find a 3NF decomposition of R with Σ .

Initial Step

Keys: $\{B,C,D\}, \{B,C,F\}, \{B,E,F\}$

Canonical Cover: $\{ \{B,E\} \rightarrow \{A,D\}, \{C\} \rightarrow \{E\}, \{D\} \rightarrow \{E\}, \{E\} \rightarrow \{C\} \}$

Construction

 $\{B,E\} \rightarrow \{A,D\}$
 $\Downarrow$
 $\{A,B,D,E\}$
 $\{C\} \rightarrow \{E\}$
 $\Downarrow$
 $\{C,E\}$
 $\{D\} \rightarrow \{E\}$
 $\Downarrow$

.. *subsumed* ..

 $\{E\} \rightarrow \{C\}$
 $\Downarrow$

.. *subsumed* ..

Case | BCNF | **3NF** 1 2 3 4 5 | Schema

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3NF DecompositionLet $R = \{A, B, C, D, E, F\}$ $\Sigma = \{ \{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\}, \{D\} \rightarrow \{C,E\}, \{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\} \}$ Find a 3NF decomposition of R with Σ .**Initial Step**Keys: $\{B,C,D\}, \{B,C,F\}, \{B,E,F\}$ Canonical Cover: $\{ \{B,E\} \rightarrow \{A,D\}, \{C\} \rightarrow \{E\}, \{D\} \rightarrow \{E\}, \{E\} \rightarrow \{C\} \}$ **Construction** $\{B,E\} \rightarrow \{A,D\}$ $\Downarrow$ $\{A,B,D,E\}$ $\{C\} \rightarrow \{E\}$ $\Downarrow$ $\{C,E\}$ $\{D\} \rightarrow \{E\}$ $\Downarrow$ *.. subsumed ..* $\{E\} \rightarrow \{C\}$ $\Downarrow$ *.. subsumed ..***Add Key (if needed)**None of the relations contained $\{B,C,F\}, \{B,D,F\}$, or $\{B,E,F\}$. We add **$\{B,C,F\}$** (*we can also add the others*).

Case | BCNF | **3NF** 1 2 3 4 5 | Schema

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Fragments

- {A,B,D,E} { {B,E} → {A,D}, {D} → {E} }
- {C,E} { {C} → {E}, {E} → {C} }
- {B,C,F} ∅

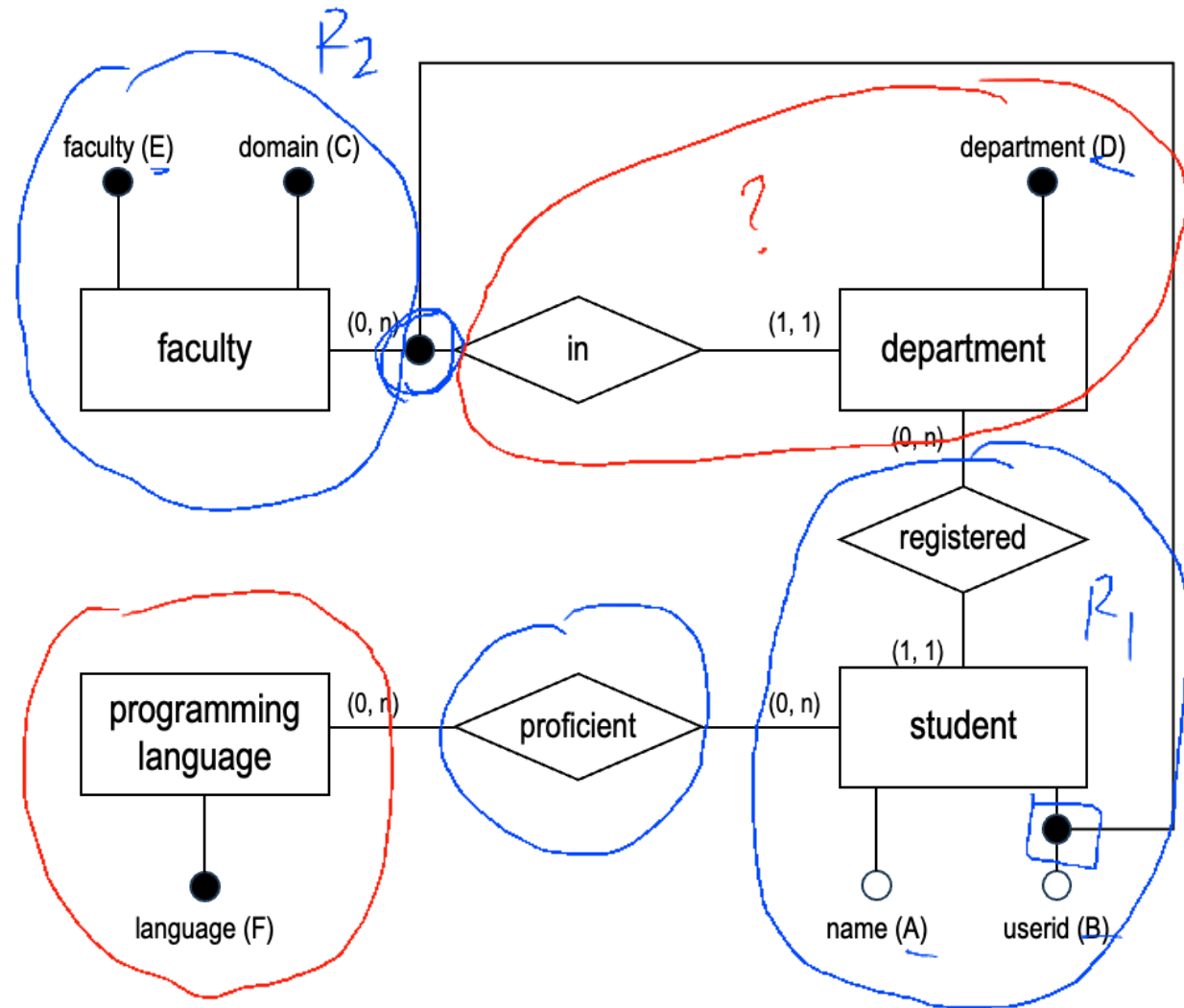
 $\Sigma = \{$

$\{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\},$
 $\{D\} \rightarrow \{C,E\},$
 $\{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\}$

}

Question

Can you map to the ERD?



Case | BCNF | **3NF** 1 2 3 4 5 | Schema

Sentosa University of Technology

Fragments

- {A,B,D,E} { {B,E} → {A,D}, {D} → {E} }
- {C,E} { {C} → {E}, {E} → {C} }
- {B,C,F} ∅
- {D,E} { {D} → {E} }

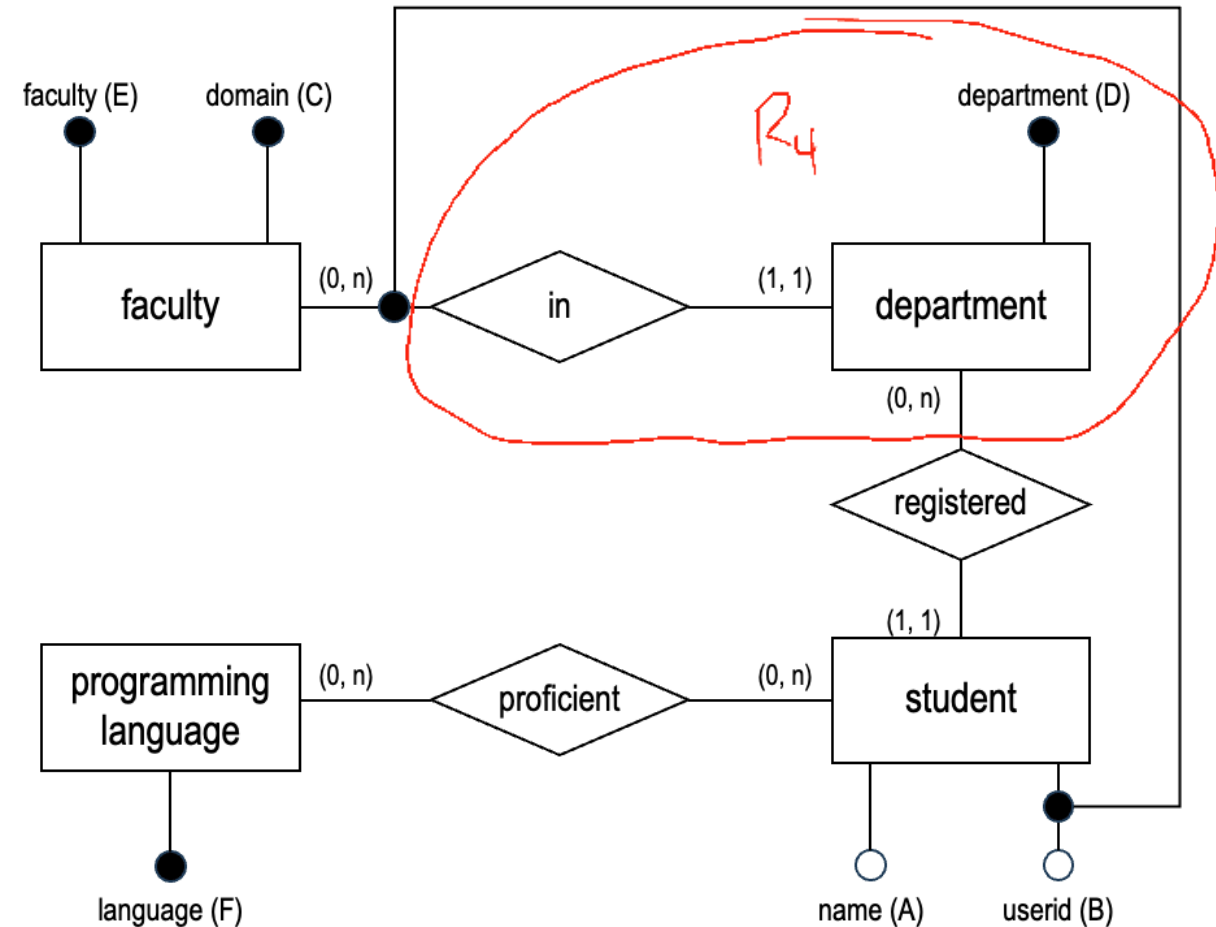
 $\Sigma = \{$

$\{C\} \rightarrow \{E\}, \{E\} \rightarrow \{C\},$
 $\{D\} \rightarrow \{C,E\},$
 $\{B,C\} \rightarrow \{A,D\}, \{B,E\} \rightarrow \{A,D\}$

}

Question

Can you map to the ERD?

What if we unsubsumed?

Case | BCNF | 3NF | **Schema** 0 2 3 3

Pathological Case

Question

Consider $R = \{A, B, D, E\}$, with $\Sigma = \{ \{B, E\} \rightarrow \{A, D\}, \{D\} \rightarrow \{E\} \}$

Enforce the functional dependency in the schema using 5 basic constraints.

Case | BCNF | 3NF | **Schema** 1 2 3 4

Pathological Case

Question

BE

BD

Consider $R = \{A, B, D, E\}$, with $\Sigma = \{ \{B, E\} \rightarrow \{A, D\}, \{D\} \rightarrow \{E\} \}$

Enforce the functional dependency in the schema using 5 basic constraints.

Subsumed

```
CREATE TABLE r1 (
  A  INT  NOT NULL,
  B  INT  NOT NULL,
  D  INT  NOT NULL,
  E  INT  NOT NULL,
  PRIMARY KEY (B, E),
  PRIMARY KEY (B, E), UNIQUE (B, D)
);
-- Adapt the datatype to the actual datatype
```

Case | BCNF | 3NF | **Schema** 1 2 3 4

Pathological Case

Question

Consider $R = \{A, B, D, E\}$, with $\Sigma = \{ \{B, E\} \rightarrow \{A, D\}, \{D\} \rightarrow \{E\} \}$

Enforce the functional dependency in the schema using 5 basic constraints.

Subsumed

```
CREATE TABLE r1 (
  A INT NOT NULL,
  B INT NOT NULL,
  D INT NOT NULL,
  E INT NOT NULL,
  PRIMARY KEY (B, E), UNIQUE (B, D)
);
-- Adapt the datatype to the actual datatype
```



A	B	D	E
1	1	1	1
2	2	1	2

Issue

The table above does not violate any primary key and unique constraint.

But it does not enforce $\{D\} \rightarrow \{E\}$.

Case | BCNF | 3NF | **Schema** 1 2 3 4

Pathological Case

Question

Consider $R = \{A, B, D, E\}$, with $\Sigma = \{ \{B, E\} \rightarrow \{A, D\}, \{D\} \rightarrow \{E\} \}$

Enforce the functional dependency in the schema using 5 basic constraints.

What if Not Subsumed?

```
CREATE TABLE r1 (  
  A INT NOT NULL,  
  B INT NOT NULL,  
  D INT NOT NULL,  
  E INT NOT NULL,  
  PRIMARY KEY (B, E), UNIQUE (B, D),  
  FOREIGN KEY (D, E) REFERENCES r4(D, E)  
);
```

$BE \rightarrow AD$
 $BD \rightarrow AE$

```
CREATE TABLE r4 (  
  D INT NOT NULL, UNIQUE  
  E INT NOT NULL UNIQUE,  
  PRIMARY KEY (D, E)  
);
```

$D \rightarrow E$

```
postgres=# exit
```

```
Press any key to continue . . .
```